S.	Z.		Stelle	Befund	Vorschlag
28	1692	◐	Beweis der Proposition in Abschnitt 1.3.8	**Umgesetzt.** Proposition 1.1.9 (ii) liefert, dass $D(A) \otimes D(B)$ ein Core für den Generator C von $(S(t) \otimes T(t))$ ist; „core of $A \otimes \mathrm{Id} + \mathrm{Id} \otimes B$“ ist trivial (das ist dessen Definitionsbereich) und gibt nicht die Behauptung. Gemeint: $C = \overline{A \otimes \mathrm{Id} + \mathrm{Id} \otimes B}$.	hence it is a core for the generator by Proposition prop:a1-1.9 (ii), i.e. the generator is the closure of $A \otimes \mathrm{Id} + \mathrm{Id} \otimes B$.
29	1704	○	Abschnitt 1.3.9 (a)	**Umgesetzt.** Mit Vertauschbarkeit nur zu gleichen Zeiten $S(t)T(t) = T(t)S(t)$ folgt weder direkt die Halbgruppeneigenschaft von $U(t) = S(t)T(t)$ (dafür braucht man $T(t)S(s) = S(s)T(t)$) noch (b) über die Laplace-Transformation. Üblich ist die Vertauschbarkeit für alle s, t.	$S(s)T(t) = T(t)S(s)$ for all $s, t \geqslant 0$

Teil I, Kap. 2 – Characterization of Semigroups on Banach Spaces (`part-a/chap-a2.tex`)

S.	Z.		Stelle	Befund	Vorschlag
33	214	●	Beweis von Theorem 2.1.1, $(a) \Rightarrow (b)$	Für $f \in D(B)$ (B Generator auf E_1) existiert $\lim_{t\to 0} \frac{1}{t}(T_1(t)f - f)$ in der Norm von E_1, nicht nur in der von E. Gebraucht wird genau das: Nur die E_1-Konvergenz liefert die Konvergenz von $A\frac{1}{t}(T_1(t)f - f)$, und erst daraus folgt mit der Abgeschlossenheit von A, dass $Af \in D(A)$. Mit „in the norm of E“ ist der Schluss in Z. 222 nicht gedeckt.	exists in the norm of E_1. ✓
33	278	◐	Beweis von Theorem 2.1.1, $(b) \Rightarrow (a)$	$A_1\dot{v}(t) = A_1u(t)$ ist nicht definiert: Bekannt ist nur $v(t) \in D(A^2)$, also $u(t) = Av(t) \in D(A)$, aber nicht $u(t) \in D(A^2) = D(A_1)$. Gemeint ist $A\dot{v}(t)$ (= $AAv(t) = A^2v(t)$).	$\frac{\mathrm{d}}{\mathrm{d}t}Av(t) = Au(t) = A\dot{v}(t) = A^2v(t)$. ✓
35	350	○	Beweis von Corollary 2.1.2, $(d) \Rightarrow (a)$	Bedingung (d) verlangt Gleichmäßigkeit nur für $t \in [0, 1]$. Daraus folgen beschränkte $T(t)$ mit $u(t, f) = T(t)f$ zunächst nur für $0 \leqslant t \leqslant 1$; für $t > 1$ wird $T(t)$ erst in Z. 371 definiert. „for all $t \geqslant 0$“ greift also vor und passt nicht zu Z. 366–371.	for all $0 \leqslant t \leqslant 1$; bitte prüfen (1)
36	409	●	Example 2.1.4	Mit $A = \begin{pmatrix} 0 & -B \\ 0 & 0 \end{pmatrix}$ ist $A(f, g) = (-Bg, 0)$, also $(\lambda - A)(f, g) = (\lambda f + Bg, \lambda g)$, nicht $(\lambda f - Bg, \lambda g)$. Am Ergebnis $\varrho(A) = \emptyset$ ändert das nichts. Z. 417 hat denselben Vorzeichenkonflikt. *Hinweis 20.10.2026:* Die Lösung in Z. 417 ist über korrekturen-mathe-part-a schon auf $(f - tBg, g)$ geändert, die Matrix ist geblieben. Hier daher $(\lambda f + Bg, \lambda g)$ setzen, die Matrix nicht ändern.	Am einfachsten in Z. 398 $A := \begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix}$ setzen; ✓ dann sind Z. 403, 409 und 417 unverändert richtig. (2) Sonst hier $(\lambda f + Bg, \lambda g)$.
36	417	●	Example 2.1.4	Ergänzt korrekturen-mathe. Nachgerechnet: Für $u(t) = (f + tBg, g)$ ist $\dot{u}(t) = (Bg, 0)$, aber $Au(t) = (-Bg, 0)$; die Lösung ist $(f - tBg, g)$. Der Fehler ist also sicher (nicht nur ○), und Z. 409 hat denselben Vorzeichenfehler. *Hinweis 20.10.2026:* Im Text schon erledigt: $u(t) = (f - tBg, g)$ (über korrekturen-mathe-part-a, Kap. 2); die Matrix bleibt unverändert.	wie bei Z. 409: Matrix in Z. 398 zu $\begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix}$ ändern ~~oder hier $u(t) = (f - tBg, g)$~~ erledigt sich durch die Korrektur (2).

Proof. It is clear that (a) implies the remaining assertions. So assume that A satisfies (EU). Then by Theorem 2.1.1, A_1 is a generator. If there exists $\lambda \in \varrho(A)$, then $(\lambda - A)$ is an isomorphism from E_1 onto E and A is similar to A_1 via this isomorphism since

$$D(A_1) = \{(\lambda - A)^{-1} f : f \in D(A)\}$$

and

$$Af = (\lambda - A)A_1(\lambda - A)^{-1} f \quad \text{for all} \quad f \in D(A)$$

(see Section 1.3.1). Thus A is a generator on E and we have shown that (b) implies (a).

If (c) holds, then there exists $\lambda > s(A_1)$ such that $(\lambda - A)D(A) = E$. We show that $(\lambda - A)$ is injective. Then $\lambda \in \varrho(A)$ since A is closed. Assume that $\lambda f = Af$ for some $f \in D(A)$. Then $f \in D(A^2) = D(A_1)$, and so $f = 0$ since $\lambda \in \varrho(A_1)$. This proves that (c) implies (b).

It remains to show that (d) implies (a). Assertion (d) implies that for all $t \geqslant 0$ there exist bounded operators $T(t) \in \mathcal{L}(E)$ such that

$$u(t, f) = T(t)f \quad \text{if} \quad f \in D(A).$$

Moreover,

$$\sup_{0 \leqslant t \leqslant 1} \|T(t)\| < \infty.$$

It follows that $T(\cdot)f$ is strongly continuous for all $f \in E$ (since it is so for $f \in D(A)$ and $D(A)$ is dense).

Let $t > 1$. There exist unique $n \in \mathbb{N}$ and $s \in [0, 1)$ such that $t = n + s$. Let

$$T(t) := T(1)^n \cdot T(s).$$

From the uniqueness of the solutions it follows that $T(t)f = u(t, f)$ for all $t \geqslant 0$ as well as $T(t + s)f = T(s)T(t)f$ for all $f \in D(A)$ and $s, t \geqslant 0$. Thus T is a semigroup.

Denote by B its generator. It follows from the definition that $A \subseteq B$. Moreover, $D(A)$ is invariant under the semigroup. So by Theorem 1.1.9 (ii) $D(A)$ is a core of B. Since A is closed this implies that $A = B$. □

Remark 2.1.3 It is surprising that from condition (b) and (c) in the corollary it follows automatically that $D(A)$ is dense. On the other hand, this condition cannot be omitted in (d). In fact, consider any generator $\tilde{A}$ and its restriction A with domain $D(A) = \{0\}$. Then A satisfies the remaining conditions in (d) but is not a generator (if $\dim E > 0$).

Example 2.1.4 We give a densely defined closed operator A such that there exists a unique solution of (ACP) for all initial values in $D(A)$, but A is not a generator.

UNCORRECTED PROOF

Let B be a densely defined unbounded closed operator on a Banach space F. Consider $E = F \oplus F$ and A on E given by

$$A := \begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix}$$

with domain $F \times D(B)$.

Then $D(A^2) = \{(f,g) \in F \times D(B) : Bg \in F\} = D(A)$ and so $A_1 \in \mathcal{L}(E_1)$. In particular, A_1 is a generator while A is not. For instance condition (b) in Corollary 2.1.2 does not hold since, for each $\lambda \in \mathbb{C}$

$$(\lambda - A)D(A) = \{(\lambda f - Bg, \lambda g) : f \in F, g \in D(B)\} \subseteq F \times D(B) \neq F \times F = E,$$

so $\varrho(A) = \emptyset$.

As a further illustration, we note that the solution of the corresponding abstract Cauchy problem for the initial value $(f,g) \in F \times D(B)$ is given by

$$u(t) = (f + tBg, g).$$

Since B is unbounded, condition (d) of Corollary 2.1.2 is clearly violated.

Remark It may happen that a generator A can be extended to a closed operator B. Then one can consider the abstract Cauchy problem (ACP) associated with B. It also has a solution for every initial value in $D(B)$, but none of the solutions is unique unless $A = B$.

In fact, denote by $(T(t))_{t \geq 0}$ the semigroup generated by A. Let $f \in D(B)$. Let $\lambda > \omega_0(A)$. Then there exists $g \in D(A)$ such that $(\lambda - B)f = (\lambda - A)g$. Let $h = f - g$. Then $h \in \ker(\lambda - B)$. Define u by

$$u(t) = \mathrm{e}^{\lambda t} h + T(t)g.$$

Then u is a solution of (ACP) for B with initial value f. It follows from Corollary 2.1.2 that there exists a non-trivial solution for the initial value 0.

2.1.2 One-parameter groups

Generators of one-parameter groups can be characterized similarly by existence and uniqueness of the solutions of the associated Cauchy problem. However, here the assumption of continuous dependence on the initial values can be relaxed (in fact, one has no longer to assume that the continuous dependence is uniform in t).

Theorem 2.1.5 *Let A be a closed densely defined operator. The following assertions are equivalent.*

UNCORRECTED PROOF

S.	Z.		Stelle	Befund	Vorschlag
37	499	●	Beweis von Theorem 2.1.6	„Since $T(\cdot)f$ is continuous, it follows that $T(\cdot)f$ is measurable“ ist zirkulär: Die Stetigkeit von $T(\cdot)f$ für beliebiges $f \in E$ soll erst bewiesen werden. Gemeint ist: $T(\cdot)f_n$ ist stetig ($f_n \in D(A)$), also ist der punktweise Limes $T(\cdot)f$ messbar.	Since $T(\cdot)f_n$ is continuous, it follows that $T(\cdot)f$ is measurable.
37	510	○	Beweis von Theorem 2.1.6	Gezeigt wird nur, dass $(T(t))$ eine stark stetige Gruppe ist (also $t \in \mathbb{R}$, nicht $t \geqslant 0$). Dass ihr Generator B gleich A ist, wird nicht gezeigt. Man schließt wie im Beweis von Corollary 2.1.2: $A \subseteq B$, $D(A)$ ist dicht und invariant, also ein Kern von B (Proposition 1.1.9 (ii)), und A ist abgeschlossen.	$(T(t))_{t\in\mathbb{R}}$; Satz ergänzen, z. B. As in the proof of Corollary cor:a2.1.2 it follows that A is the generator.; bitte prüfen
39	599	●	Definition des Sektors $S(\alpha)$ (vor Def. 2.1.11)	Mit $r \geqslant 0$ liegt $0 \in S(\alpha)$. Dann verlangt Theorem 2.1.12 $0 \in S(\alpha + \pi/2) \subseteq \varrho(A)$, und $\|R(\lambda, A)\| \leqslant M/|\lambda|$ ist in $\lambda = 0$ sinnlos. Das widerspricht Corollary 2.1.13 ($A = d/dx$ auf $L^2(\mathbb{R})$, A^2 = Laplace, $0 \in \sigma(A^2)$) und betrifft auch Example 2.1.28 (iii) (Z. 1468, 1473).	$r > 0$ (die Einschränkung auf $[0, \infty)$ in Def. 2.1.11 dann mit $T(0) = \mathrm{Id}$ lesen); bitte prüfen
40	689	●	Beweis von Corollary 2.1.13, Gl. (1.7)	Aus $\lambda - A^2 = (re^{i\beta} - A)(re^{i\beta} + A)$ und $re^{i\beta} + A = re^{i\beta} - (-A)$ folgt $R(\lambda, A^2) = R(re^{i\beta}, A)R(re^{i\beta}, -A)$; im zweiten Faktor steht fälschlich $e^{-i\beta}$. Die Abschätzung danach bleibt gültig, weil $\mathrm{Re}(re^{\pm i\beta}) = r\cos\beta$. Vgl. den richtigen analogen Ausdruck in Z. 957.	$R(\lambda, A^2) = R(re^{i\beta}, A)R(re^{i\beta}, -A)$
40	703	◐	Beweis von Corollary 2.1.13	Das zweite „$\leqslant$“ ist eine Gleichheit, und $M = (N/\cos\beta)^2$ hängt über β von λ ab. Gebraucht wird aber eine von λ unabhängige Konstante. Wegen $|\beta| < \beta_1 < \pi/2$ gilt $\cos\beta \geqslant \cos\beta_1 > 0$.	$\leqslant \frac{1}{r^2}\left(\frac{N}{\cos\beta_1}\right)^2 = M\,|\lambda|^{-1}$ mit $M := (N/\cos\beta_1)^2$
42	853	◐	Beweis von Theorem 2.1.14	Die Abschätzung durch $N|\xi + i\eta|^{-1}$ ist nur für $|\eta| \geqslant r$ gezeigt, und die Voraussetzung gilt nur für $|\lambda| \geqslant r$. Daraus folgt $\lambda \in \varrho(A)$ mit $\|R(\lambda, A)\| \leqslant N'/|\lambda|$ nur außerhalb einer Kugel, nicht auf ganz $S(\alpha + \pi/2)$. Für das Kriterium in Z. 814–822 genügt das auch.	for all $\lambda \in S(\alpha + \pi/2) \setminus B(r')$ (mit geeignetem $r' \geqslant r$)
43	943	●	Beweis von Theorem 2.1.15	Mit $\varepsilon = \cot\beta$ gilt $2\varepsilon/(\varepsilon^2 - 1) = \tan(2\beta)$. Für die Inklusion in Z. 915 braucht man $2\beta > \alpha + \pi/2$. Beide Winkel liegen in $(\pi/2, \pi)$, wo tan wächst; also ist das äquivalent zu $\tan(\alpha + \pi/2) < 2\varepsilon/(\varepsilon^2 - 1)$. Mit „$>$“ folgt $2\beta < \alpha + \pi/2$, und die Inklusion wäre falsch.	$\tan(\alpha + \pi/2) < 2\varepsilon/(\varepsilon^2 - 1)$ (d. h. $\beta \in ((\alpha + \pi/2)/2, \pi/2)$)
45	1047	○	Verweis nach Theorem 2.1.18	In Pazy (1983) stehen die differenzierbaren Halbgruppen meines Wissens in Chapter 2, Section 2.4 (Spectral properties and regularity). Chapter 3 behandelt Störungen und Approximationen.	[Chapter 2, Theorem 4.7 and 4.8]; bitte prüfen (auch die Theoremnummern)
46	1180	◐	Beweis von Theorem 2.1.20	Im Exponenten steht ω, im Beweis heißt die Wachstumskonstante aber w (Z. 1129–1134). Ein ω ist hier nicht definiert.	$Me^{(w-a)t}$
48	1209	◐	Text vor Definition 2.1.22 (kompakte Halbgruppen)	$\lim T(h+t)f = \lim T(h)T(t)f$ ist trivial (Halbgruppeneigenschaft). Zu zeigen ist die gleichmäßige Konvergenz gegen $T(t)f$ für $f \in U$. Sie folgt aus der gleichmäßigen Konvergenz auf der kompakten Menge $\overline{T(t)U}$; $T(t)(U)$ selbst ist nur relativ kompakt (Z. 1205).	$\lim_{h\downarrow 0} T(h+t)f = \lim_{h\downarrow 0} T(h)T(t)f = T(t)f$; Z. 1205 $K = \overline{T(t)(U)}$

(a) *A is the generator of a strongly continuous one-parameter group.*

(b) *For every $f \in D(A)$ there exists a unique function $u(\cdot, f) \in C^1(\mathbb{R})$ with $u(t, f) \in D(A)$ for all $t \in \mathbb{R}$, $u(0, f) = f$, and*

$$\frac{\mathrm{d}}{\mathrm{d}t} u(t, f) = Au(t, f) \qquad (t \in \mathbb{R}).$$

Moreover, if $f_n \in D(A)$ with $\lim_{n\to\infty} f_n = 0$, then $\lim_{n\to\infty} u(t, f_n) = 0$ for all $t \in \mathbb{R}$.

Proof. It is clear that (a) implies (b). If (b) holds then there exist operators $T(t) \in \mathcal{L}(E)$ such that

$$u(t, f) = T(t)f \quad (t \in \mathbb{R}, f \in D(A)).$$

It follows from the uniqueness of the solutions that $T(t + s) = T(t)T(s)$ $(s, t \in \mathbb{R})$.

Let $f \in E$. There exist $(f_n) \in D(A)$ such that $\lim_{n\to\infty} f_n = f$. Then $\lim_{n\to\infty} T(t)f_n = T(t)f$ for all $t \in \mathbb{R}$. Since $T(\cdot)f$ is continuous, it follows that $T(\cdot)f$ is measureable. Hence by Hille & Phillips [21, 10.2.1]

$$\sup_{t\in J} \|T(t)\| < \infty$$

for every compact interval $J \subseteq (0, \infty)$.

By the group property this implies that $T(\cdot)$ is norm bounded on bounded subsets of $\mathbb{R}$ and $T(\cdot)f$ is continuous if $f \in D(A)$. Since $D(A)$ is dense, this implies the strong continuity of $(T(t))_{t\geqslant 0}$. As in the proof of Corollary 2. □ 1.2 one shows that A is its generator.

2.1.3 The Hille-Yosida Theorem

Given an operator A, frequently it is easier to obtain information about its resolvent than to solve the Cauchy problem. Therefore the following theorem is central in the theory of one-parameter semigroups.

Theorem 2.1.6 (Hille-Yosida) *Let A be an operator on a Banach space E. The following conditions are equivalent.*

(a) *A is the generator of a strongly continuous semigroup.*

(b) *The domain $D(A)$ of A is dense in E and there exist $w, M \in \mathbb{R}$ such that $(w, \infty) \subseteq \varrho(A)$ and*

$$\|(\lambda - w)^n (\lambda - A)^{-n}\| \leqslant M$$

for all $\lambda > w$ and $n \in \mathbb{N}$.

UNCORRECTED PROOF

In general it is not easy to give an estimate for the powers of the resolvent in order to apply Theorem 2.1.6. However, there is an important case when it suffices to consider merely the resolvent.

Corollary 2.1.7 *For an operator A on a Banach space E the following assertions are equivalent.*

(a) *A is the generator of a strongly continuous contraction semigroup.*

(b) *A is densely defined, $(0,\infty) \subseteq \varrho(A)$ and $\|\lambda R(\lambda, A)\| \leqslant 1$ for all $\lambda > 0$.*

The difficult part in the proof of Theorem 2.1.6 is to show that (b) implies (a). One has to construct the semigroup out of the resolvent. We mention two formulas which can be used for the proof.

Proposition 2.1.8 *Let A be the generator of a strongly continuous semigroup $(T(t))_{t\geqslant 0}$. For $\lambda > 0$ let*

$$A(\lambda) = \lambda^2 R(\lambda, A) - \lambda \operatorname{Id} = \lambda A R(\lambda, A).$$

Then

$$T(t)f = \lim_{\lambda\to\infty} \mathrm{e}^{tA(\lambda)} f \tag{2.1.2}$$

for all $f \in E$ and $t \geqslant 0$.

Yosida's proof consists in showing that the limit in Eq. (2.1.2) exists under the hypothesis (b) of Theorem 2.1.6 (see Davies [11], Goldstein [15] or Pazy [31]).

The proof of Hille (see Kato [22]) is inspired by the following formula.

Proposition 2.1.9 *Let A be the generator of a strongly continuous semigroup $(T(t))_{t\geqslant 0}$. Then*

$$T(t)f = \lim_{n\to\infty} (\operatorname{Id} - t/nA)^{-n} f = \lim_{n\to\infty} (n/t \cdot R(n/t, A))^n f \tag{2.1.3}$$

for all $f \in E$ and $t \geqslant 0$.

2.1.4 Holomorphic semigroups

We now describe a hierarchy of smoothness conditions on the semigroup, starting with the most restrictive class; namely, holomorphic semigroups. The generators of these semigroups can be characterized by a particularly simple condition.

For $\alpha \in (0, \pi]$ we define the sector $S(\alpha)$ in the complex plane by

$$S(\alpha) = \{r\mathrm{e}^{\mathrm{i}\vartheta} : r \geqslant 0, \vartheta \in (-\alpha, \alpha)\}.$$

UNCORRECTED PROOF

Definition 2.1.10 Let $\alpha \in (0, \pi/2]$. A strongly continuous semigroup $(T(t))_{t\geqslant 0}$ is called a bounded holomorphic semigroup of angle α if $T(\cdot)$ is the restriction of a holomorphic function

$$T\colon S(\alpha) \to \mathcal{L}(E)$$

satisfying

(i) $T(z)T(z') = T(z+z') \quad (z \text{ and } z' \in S(\alpha))$,

(ii) For each $\alpha_1 \in (0,\alpha)$ the set $\{T(z)\colon z \in S(\alpha_1)\}$ is uniformly bounded, and $\lim_{n\to\infty} T(z_n)f = f$ for every null-sequence (z_n) in $S(\alpha_1)$ and every $f \in E$.

Remark A function $T\colon S(\alpha) \to \mathcal{L}(E)$ is holomorphic with respect to the operator norm if and only if it is strongly holomorphic if and only if it is weakly holomorphic Yosida [40, V.3].

Theorem 2.1.11 *Let A be a densely defined operator on a Banach space E and $\alpha \in (0, \pi/2]$. Then A is the generator of a bounded holomorphic semigroup of angle α if and only if*

$$S(\alpha + \pi/2) \subseteq \varrho(A)$$

and for every $\alpha_1 \in (0,\alpha)$ there exists a constant M such that

$$\|R(\lambda, A)\| \leqslant M/|\lambda| \quad (\lambda \in S(\alpha_1 + \pi/2). \tag{2.1.4}$$

For the proof we refer to Davies [11], for example.

Remark Let A be the generator of a bounded holomorphic semigroup $(T(t))_{t\geqslant 0}$ of angle α, and let $z_0 \in S(\alpha)$. Then $z_0 A$ generates a bounded semigroup $(S(t))_{t\geqslant 0}$ given by $S(t) = T(z_0 t)$ $(t \geqslant 0)$ (where again we denote by $T(z)$ the holomorphic extension of $(T(t))_{t\geqslant 0}$ on $S(\alpha)$).

As an application of Theorem 2.1.11 we prove the following.

Corollary 2.1.12 *Let A be the generator of a bounded group. Then A^2 generates a bounded holomorphic semigroup of angle $\pi/2$.*

Proof. Let $0 < \alpha_1 < \pi/2$ and $\lambda \in S(\alpha_1 + \pi/2)$. There exists $r > 0$ and $\beta \in (-\beta_1, \beta_1)$, where $\beta_1 := (\alpha_1 + \pi/2)/2$ such that $\lambda = r^2 e^{i2\beta}$. Then

$$(\lambda - A^2) = (re^{i\beta} - A)(re^{i\beta} + A)\,.$$

It follows that $\lambda \in \varrho(A)$ and

$$R(\lambda, A^2) = R(re^{i\beta}, A)R(re^{-i\beta}, -A)\,. \tag{2.1.5}$$

UNCORRECTED PROOF

exists and

$$\begin{aligned}\|R(\xi+\mathrm{i}\eta,A)\| &\leqslant (|\xi+\mathrm{i}\eta|)^{-1}\cdot|\xi+\mathrm{i}\eta|\cdot\sum_{n=0}^{\infty}|\xi|^n M^{n+1}/|\eta|^{n+1}\\ &\leqslant (|\xi+\mathrm{i}\eta|)^{-1}\cdot 2M(|\xi|^2+|\eta|^2)^{1/2}/|\eta|\cdot\sum_{n=0}^{\infty}M^n c^n\\ &\leqslant (4M\cdot(c^2+1)^{1/2})/|\xi+\mathrm{i}\eta|\\ &\leqslant N|\xi+\mathrm{i}\eta|^{-1}.\end{aligned}$$

This together with the assumption implies that there exist $N'\geqslant 0$ and $\alpha\in\,]0,\pi/2]$ such that $\lambda\in\varrho(A)$ and $\|R(\lambda,A)\|\leqslant N'/|\lambda|$ for all $\lambda\in S(\alpha+\pi/2)$. □

Compared with the Hille-Yosida theorem, Theorem 2.1.13 gives a very simple criterion for an operator to be the generator of a (holomorphic) semigroup. Merely the resolvent and not its powers have to be estimated. However, the resolvent has to be known in a right half-plane instead of a right half-line.

On the other hand, given a strongly continuous semigroup, merely an estimate on a vertical line implies that the semigroup is holomorphic. More precisely, the following holds.

Corollary *A strongly continuous semigroup with generator A is holomorphic if and only if there exist $w>\omega_0(A)$ and $M\geqslant 0$ such that one has*

$$\|R(w+\mathrm{i}\eta,A)\|\leqslant\frac{M}{|\eta|}\ \text{ for all }\ \eta\in\mathbb{R}.$$

Proof. In fact, assume that the condition holds. Since $A-w$ is the generator of a bounded semigroup, one has

$$\|R(\lambda,A-w)\|\leqslant N/\mathrm{Re}(\lambda)$$

for some $N>0$ and all $\lambda\in\mathbb{C}$ satisfying $\mathrm{Re}(\lambda)>0$.

Consequently, for every $\alpha\in(0,\pi/2)$,

$$\|R(\lambda,A-w)\|\leqslant(|\lambda|/\mathrm{Re}(\lambda))N/|\lambda|\leqslant N(\cos\alpha)^{-1}/|\lambda|\ \text{ for all }\ \lambda\in S(\alpha).$$

The remaining estimate for a sector symmetric to the imaginary axis is given by the proof of Theorem 2.1.13 and shows that $A-w$ generates a holomorphic semigroup. The reverse implication is clear. □

We now prove the following extension of Corollary 2.1.12

Theorem 2.1.14 *Let A be the generator of a strongly continuous group. Then A^2 generates a holomorphic semigroup of angle $\pi/2$.*

UNCORRECTED PROOF

Proof. There exists $w \geqslant 0$ such that $(\pm A - w)$ generates a bounded semigroup. Consequently, there exists $M \geqslant 0$ such that

$$\|R(\mu, \pm A - w)\| \leqslant M/\mathrm{Re}(\mu)$$

whenever $\mathrm{Re}(\mu) > 0$.

Let $\alpha \in (0, \pi/2)$. There exist $r_0 \geqslant 0$ and $\beta \in (0, \pi/2)$ such that

$$S(\alpha + \pi/2) \setminus B(r_0) \subseteq \{z^2 \colon z \in S(\beta) + w\}.$$

In fact, the line

$$\{w + r(\cos\beta + \mathrm{i}\sin\beta) \colon r \geqslant 0\}$$

can be parameterized by

$$z(t) = w + t + \mathrm{i} \cdot t/\varepsilon \quad (t \geqslant 0),$$

where $\varepsilon > 0$ depends on β. Then

$$z(t)^2 = (w + t)^2 - t^2/\varepsilon^2 + \mathrm{i}2t(w + t)/\varepsilon.$$

Thus

$$\lim_{t\to\infty} \mathrm{Im}\{z(t)\}^2/\mathrm{Re}\{z(t)\}^2 = 2\varepsilon/(\varepsilon^2 - 1).$$

Choose $\beta \in (\pi/4, \pi/2)$ such that $\tan(\alpha + \pi/2) > 2\varepsilon/(\varepsilon^2 - 1)$.

Now let $\lambda \in S(\alpha + \pi/2) \setminus B(r_0)$. Then there exist $\vartheta \in (-\beta, \beta)$ and $r \geqslant 0$ such that

$$\lambda = (r\mathrm{e}^{\mathrm{i}\vartheta} + \omega)^2.$$

Thus

$$(\lambda - A^2) = (r\mathrm{e}^{\mathrm{i}\vartheta} + w - A)(r\mathrm{e}^{\mathrm{i}\vartheta} + w + A).$$

Hence $\lambda \in \varrho(A^2)$ and $R(\lambda, A^2) = R(r\mathrm{e}^{\mathrm{i}\vartheta}, A - w)R(r\mathrm{e}^{\mathrm{i}\vartheta}, -A - w)$. We conclude that

$$|\lambda| \cdot \|R(\lambda, A^2)\| \leqslant |\lambda| \cdot M^2/(\cos\vartheta)^2 r^2 \leqslant (|\lambda|/r^2) \cdot M^2/(\cos\beta)^2.$$

Thus $|\lambda| \cdot \|R(\lambda, A^2)\|$ is uniformly bounded for $\lambda \in S(\alpha + \pi/2) \setminus B(r_0)$. □

Remark In Theorem 2.1.14 the assumption that $\pm A$ are generators can be relaxed. In fact, the proof shows the following. If A is a densely defined operator such that $\{\lambda \in \mathbb{C} \colon \mathrm{Re}(\lambda) > 0\} \subseteq \varrho(\pm A - w)$ and $\|R(\lambda, \pm A - w)\| \leqslant M/\mathrm{Re}(\lambda)$ for some $M \geqslant 0$, $w \geqslant 0$, then A^2 generates a holomorphic semigroup of angle $\pi/2$.

UNCORRECTED PROOF

S.	Z.		Stelle	Befund	Vorschlag
49	1280	○	Beweis von Theorem 2.1.25	$s \mapsto \mathrm{e}^{-ws}T(s)$ ist nur auf $(0,\infty)$ normstetig, nicht in $s = 0$ ($T(0) = \mathrm{Id}$ ist i. A. nicht kompakt). Riemann-Summen über $[0,t]$ konvergieren deshalb nicht ohne Weiteres in der Operatornorm. Man muss $\int_\delta^t$ nehmen und $\| \int_0^\delta \| \leqslant \delta \sup_{s\leqslant t} \|T(s)\|$ abschätzen.	Begründung kurz ergänzen ($\int_\delta^t$ kompakt, $\delta \to 0$); bitte prüfen (7)
50	1359	●	Example 2.1.28	$m\colon X \to \mathbb{R}$ passt nicht. Im Beispiel wird $\mathrm{Re}(m(x))$ benutzt, $\sigma(A)$ liegt in $\mathbb{C}$, und die konkreten Beispiele (Z. 1509–1517: $-n + \mathrm{i}\exp(n^2)$ usw.) sind komplexwertig. Für reelles m wären (ii)–(iv) und die Beispiele gegenstandslos.	$m\colon X \to \mathbb{C}$ ✓ Bei mir ist das Bsp. 2.1.27
50	1392	●	Beweis von Example 2.1.28 (i)	Die Abschätzung $\mathrm{e}^{t\,\mathrm{Re}(m(x))} \leqslant \mathrm{e}^{(t'+1)b}$ für $\mathrm{Re}(m(x)) \leqslant b$ und $\lvert t-t'\rvert \leqslant 1$ (Z. 1397/1400) ist falsch. Es ist $b < 0$, also $tb \geqslant (t'+1)b$; die Ungleichung geht in die falsche Richtung. Für $t' < 1$ darf t sogar nahe 0 sein, dann ist $\mathrm{e}^{t\,\mathrm{Re}(m(x))}$ nahe 1.	t von 0 weghalten: $b < 0$ mit $2\mathrm{e}^{t'b/2} < \varepsilon$ wählen; dann gilt $\lvert \mathrm{e}^{tm(x)} - \mathrm{e}^{t'm(x)}\rvert \leqslant 2\mathrm{e}^{t'b/2} < \varepsilon$ für $\lvert t-t'\rvert \leqslant t'/2$; entsprechend in Z. 1417 $\delta \in (0, t'/2]$ (8) (9) ✓
53	1613	●	Beweis von Theorem 2.1.30	Aus (1.12) folgt $\|R(\lambda_0+\mathrm{i}\eta, A+B)\| \leqslant \|R(\lambda_0+\mathrm{i}\eta, A)\|\,\|(\mathrm{Id}-BR(\lambda_0+\mathrm{i}\eta, A))^{-1}\| \leqslant 2M/\lvert\eta\rvert$. Die Zeile schreibt $\|(\mathrm{Id}-BR)^{-1}\| \cdot 2M/\lvert\eta\rvert \leqslant 2M/\lvert\eta\rvert$. Das setzt $\|(\mathrm{Id}-BR)^{-1}\| \leqslant 1$ voraus, bekannt ist aber nur $\leqslant 2$ (Neumann-Reihe mit $\|BR\| \leqslant 1/2$).	$\leqslant \|(\mathrm{Id}-BR(\lambda_0+\mathrm{i}\eta, A))^{-1}\| \cdot M/\lvert\eta\rvert \leqslant 2M/\lvert\eta\rvert$ ✓ (10)
55	1720	●	Beweis von Corollary 2.1.32	Im Beweis von Theorem 2.1.31 gilt $U(A+C)U^{-1} = A+B$. Dann ist $S(t) = U\mathrm{e}^{t(A+C)}U^{-1}$, also hat $(U^{-1}S(t)U)_{t\geqslant 0}$ den Generator A+C und nicht $(US(t)U^{-1})_{t\geqslant 0}$: Die Rollen von U und U^{-1} sind vertauscht.	because $(U^{-1}S(t)U)_{t\geqslant 0}$ has $A+C$ as generator ✓ (11)
57	1829	●	Remark nach der Definition „p-contractive“ (Abschnitt 2.2)	Mit $N^+(f) = \sup_x f(x)$ ist die Aussage „N^+-kontraktiv $\Leftrightarrow$ positiv und $\|T\| \leqslant 1$“ falsch. Gegenbeispiel: $T = 0$ mit $f = -1$ ergibt $N^+(Tf) = 0 > -1 = N^+(f)$. Bei dieser Definition ist N^+-kontraktiv äquivalent zu „positiv und $T1 = 1$“. Gemeint ist die kanonische Halbnorm $\|f^+\|$ (vgl. chap-c2 Z. 49).	$N^+(f) = \sup_{x\in[0,1]} f^+(x) = \|f^+\|_\infty$ ✓ (12)
58	1904	●	Proposition 2.2.3, Gl. (2.8)	Für $w < 0$ ist die Äquivalenz falsch. Beispiel: $A = -2\,\mathrm{Id}$ auf $\mathbb{R}$, $p = N$, $w = -2$; A ist dissipativ, aber für $\lambda = -3/2$ gilt $\lambda R(\lambda, A) = -3$. Aus (2.7) bekommt man (2.8) nur mit $t = 1/\lambda$, also für $\lambda > 0$. Dasselbe gilt im Beweis von Theorem 2.2.6 (Z. 2024, 2027) für $\lambda > \omega_0(A)$, falls $\omega_0(A) < 0$.	for some $w \geqslant 0$ (bzw. $\lambda > \max\{w, 0\}$); in Z. 2024/2027 $\lambda > \max\{\omega_0(A), 0\}$ ✓ (13)
60	2101	●	Remark 2.2.8	$E_p = \{f\colon p(-f) \geqslant 0\}$ ist i. A. kein echter Kegel. Für die kanonische Halbnorm $p(f) = \mathrm{dist}(-f, E_+) \geqslant 0$ wäre $E_p = E$, im Widerspruch zu $E_+ = E_p$ (Z. 2113). Mit $p(-f) \leqslant 0$ stimmt alles: E_p ist abgeschlossen, ein Kegel (Sublinearität), echt (wegen (2.11)), p-kontraktive Operatoren lassen ihn invariant, und für die kanonische Halbnorm ist $E_p = E_+$.	$E_p := \{f \in E\colon p(-f) \leqslant 0\}$ ✓ (14)

Teil I, Kap. 3 – Spectral Theory (`part-a/chap-a3.tex`)

for all $f \in E$, and so

$$S(t) = R(0, A)(\mathrm{Id} - T(t))$$

is compact for all $t \geqslant 0$.

Since $t \mapsto T(t)$ is norm continuous for $t \geqslant t_0$, one has

$$\lim_{h \downarrow 0} \frac{1}{h}(S(t_0 + h) - S(t_0)) = T(t_0)$$

in the operator norm. Thus $T(t_0)$ is compact as limit of compact operators. □

Theorem 2.1.24 *A strongly continuous semigroup $(T(t))_{t \geqslant 0}$ is compact if and only if it is norm continuous and its generator A has compact resolvent.*

Proof. Assume that $(T(t))_{t \geqslant 0}$ is compact. Then $T(\cdot)$ is norm continuous on $(0, \infty)$, and so

$$\int_0^t \mathrm{e}^{-ws} T(s)\, \mathrm{d}s$$

is compact as the norm limit of linear combinations of compact operators, whenever $w > \omega_0(A)$. Since

$$R(w, A) = \lim_{t \to \infty} \int_0^t \mathrm{e}^{-ws} T(s)\, \mathrm{d}s$$

in the operator norm, it follows that $R(w, A)$ is compact. This proves one implication. The other follows from Proposition 2.1.23. □

Remark 2.1.25 (i) Generators of eventually compact semigroups do not necessarily have compact resolvent. Consider the nilpotent translation semigroup $(T(t))_{t \geqslant 0}$ on $F := L^1([0, 1])$ (see Section 1.2.6). Let

$$E = F \widetilde{\otimes}_\pi F = L^1([0, 1] \times [0, 1])$$

and

$$S(t) = T(t) \otimes \mathrm{Id} \quad (t \geqslant 0).$$

Then $(S(t))_{t \geqslant 0}$ is a strongly continuous semigroup with B as its generator (see Section 1.3.8). Clearly $(S(t))_{t \geqslant 0}$ is a nilpotent semigroup (so it is eventually compact), but

$$R(\lambda, B) = R(\lambda, A) \otimes \mathrm{Id}$$

is not compact.

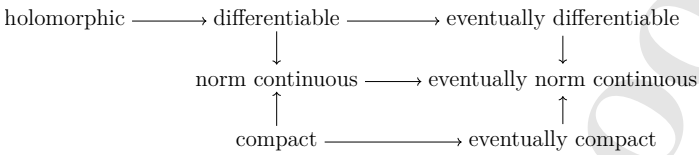
(ii) It is obvious that a group $(T(t))_{t \in \mathbb{R}}$ is eventually norm continuous if and only if it is norm continuous at 0, i.e., its generator is bounded.

On the other hand, the generator of the rotation group (Section 1.2.5) has a compact resolvent. Hence this condition does not imply any smoothness property of the semigroup.

UNCORRECTED PROOF

Positive eventually compact semigroups have remarkable properties in the setting of the Perron-Frobenius theory (see e.g., Part-II, Corollary 3.2.12).

The following scheme indicates the relation between the different classes of semigroups defined so far.



All these classes are different. This is shown by the following examples.

Example 2.1.26 The nilpotent shift semigroup (Section 1.2.6) is obviously eventually differentiable, eventually compact and eventually norm continuous. But it is not norm continuous and consequently not differentiable or compact.

Example 2.1.27 We consider multiplication semigroups (see Section 1.2.3). Let $E = C_0(X)$, where X is a locally compact space, or $E = L^p(X, \Sigma, \mu)$, where $1 \leqslant p < \infty$ and (X, Σ, μ) is a σ-finite measure space. Let $m\colon X \to \mathbb{R}$ be continuous (resp. measurable) such that $[\text{ess}]\text{-}\sup_{x\in X} \operatorname{Re}(m(x)) < \infty$.

Then $Af = m \cdot f$ with domain $D(A) = \{f \in E\colon m \cdot f \in E\}$ is the generator of the semigroup $(T(t))_{t\geqslant 0}$ given by

$$(T(t)f)(x) = \mathrm{e}^{tm(x)} f(x), \quad (t \geqslant 0, x \in X, f \in E)\,.$$

Observe that $\sigma(A) = m(X)^-$ in case $E = C_0(X)$ and

$$\sigma(A) = [\text{ess}]\text{-image}(m) := \{\lambda \in \mathbb{C}\colon \mu(\{x \in X\colon |m(x) - \lambda| < \varepsilon\}) \neq 0 \text{ for all } \varepsilon > 0\}$$

if $E = L^p$ (see Section 1.2.3). Consequently,

$$s(A) = \omega_0(A) = [\text{ess}]\text{-}\sup_{x\in X} \operatorname{Re}(m(x)).$$

(i) *The semigroup is norm continuous for $t > 0$ if and only if it is eventually norm continuous if and only if $\{\lambda \in \sigma(A)\colon \operatorname{Re}(\lambda) \geqslant b\}$ is bounded for every $b \in \mathbb{R}$. Thus the property proved in Theorem 2.1.19 characterizes generators of eventually norm continuous semigroups in the case that the semigroup consists of multiplication operators.*

Proof. Assume that $\{\lambda \in \sigma(A)\colon \operatorname{Re}(\lambda) \geqslant b\}$ is bounded for every $b \in \mathbb{R}$. Let $t' > 0$. We show that the semigroup is norm continuous at t'. Take $\varepsilon > 0$ and $b \in \mathbb{R}$ such that $2\mathrm{e}^{(t'+1)b} < \varepsilon$.

If $\operatorname{Re}(m(x)) \leqslant b$, then

$$\left|\mathrm{e}^{tm(x)} - \mathrm{e}^{t'm(x)}\right| \leqslant \mathrm{e}^{t\operatorname{Re}(m(x))} + \mathrm{e}^{t'\operatorname{Re}(m(x))} \leqslant 2\mathrm{e}^{(t'+1)b} < \varepsilon$$

UNCORRECTED PROOF

whenever $|t - t'| \leqslant 1$.

By hypothesis, the set

$$H := \{m(x) \colon x \in X, \operatorname{Re}(m(x)) \geqslant b\}$$

in the case $E = C_0(X)$ and

$$H := \{m(x) \colon \operatorname{Re}(\lambda) \geqslant b \text{ and for all } \eta > 0, \mu(\{x \in X \colon |m(x) - \lambda| < \eta\}) \neq 0\}$$

in the case $E = L^p$ is a bounded subset of $\mathbb{C}$. Thus $\lim_{t \to t'} |\mathrm{e}^{tz} - \mathrm{e}^{t'z}| = 0$ uniformly for $z \in H$. Hence there exists $\delta \in]0, 1]$ such that

$$[\text{ess}]\text{-}\operatorname{ess\,sup}\left\{\left|\mathrm{e}^{tm(x)} - \mathrm{e}^{t'm(x)}\right| \colon x \in X, \operatorname{Re}(m(x)) > b\right\} < \varepsilon$$

whenever $|t - t'| < \delta$. Together with the inequality above, we obtain that

$$\|T(t) - T(t')\| = [\text{ess}]\text{-}\sup\left\{\left|\mathrm{e}^{tm(x)} - \mathrm{e}^{t'm(x)}\right| \colon x \in X\right\} < \varepsilon$$

whenever $|t - t'| < \delta$. We have shown that the semigroup is norm continuous for $t > 0$ whenever $\{\lambda \in \sigma(A) \colon \operatorname{Re}(\lambda) \geqslant b\}$ is bounded for all $b \in \mathbb{R}$. □

(ii) *The semigroup is right-sided differentiable at a point $t > 0$ if and only if there exists $c > 0$ such that*

$$\{\lambda \in \mathbb{C} \colon |\operatorname{Im}(\lambda)| > c \cdot \mathrm{e}^{-t \operatorname{Re}(\lambda)}\} \subseteq \varrho(A).$$

Proof. The semigroup is right-sided differentiable at a point $t > 0$ if and only if the following equivalent assertions holds. (a) $T(t)E \subseteq D(A)$, (b) $\mathrm{e}^{tm} \cdot f \cdot m \in E$ for all $f \in E$, (c) $\mathrm{e}^{tm} \cdot m$ is [essentially] bounded, (d) $\mathrm{e}^{t \operatorname{Re}(m)} \cdot \operatorname{Im}(m)$ is [essentially] bounded, (e) there exists $c > 0$ such that $[\text{ess}]\text{-}\operatorname{im}(m) \subseteq \{\lambda \in \mathbb{C} \colon \mathrm{e}^{t \operatorname{Re}(\lambda)} |\operatorname{Im}(\lambda)| \leqslant c\}$, (f) there exists $c > 0$ such that $\{\lambda \in \mathbb{C} \colon |\operatorname{Im}(\lambda)| > c \cdot \mathrm{e}^{-t \operatorname{Re}(\lambda)}\} \subseteq \varrho(A)$. □

(iii) $(T(t))_{t \geqslant 0}$ *is a bounded holomorphic semigroup of angle ϑ if and only if*

$$S(\vartheta + \pi/2) \subseteq \varrho(A).$$

Proof. The condition is necessary by Theorem 2.1.11. Conversely, if $S(\vartheta + \pi/2) \subset \varrho(A)$, then one verifies directly that

$$(T(z)f)(x) = \mathrm{e}^{z \cdot m(x)} f(x) \quad (f \in E, x \in X)$$

defines a family $(T(z))_{z \in S(\vartheta)}$ of bounded operators satisfying conditions (i) and (ii) of Definition 2.1.10. □

(iv) *Choosing $X = \mathbb{N}$ and the counting measure we have $E = c_0$ or ℓ^p. Then A has a compact resolvent if*

$$\lim_{n \to \infty} |m(n)| = \infty.$$

UNCORRECTED PROOF

where the series converges in the operator norm uniformly on bounded intervals. We refer to Davies [11, III.1], Goldstein [16, I.6] or Pazy [31, Chapter 3] for these results.

Several special properties discussed above are preserved by bounded perturbations.

Theorem 2.1.29 *Let $(T(t))_{t\geqslant 0}$ be a strongly continuous semigroup with generator A, and let $B \in \mathcal{L}(E)$.*

(i) *If $(T(t))_{t\geqslant 0}$ is holomorphic or norm continuous or compact, then so is the semigroup $(S(t))_{t\geqslant 0}$ generated by $A + B$.*

(ii) *If A has a compact resolvent then so has $A + B$.*

(iii) *Let $t_0 \geqslant 0$. If $(T(t))_{t\geqslant 0}$ is norm continuous for $t > t_0$ and if B is compact, then $(S(t))_{t\geqslant 0}$ is also norm continuous for $t > t_0$.*

Proof. If $(T(t))_{t\geqslant 0}$ is norm continuous for $t > 0$, then $S_n(t)$ in Eq. (2.1.8) on the preceding page is norm continuous at $t > 0$ for every n. Thus $(S(t))_{t\geqslant 0}$ is norm continuous at $t > 0$ by Eq. (2.1.9) on the facing page. There exists $\lambda_0 \in \mathbb{R}$ such that

$$\|R(\lambda, A)\| \leqslant (2\|B\|)^{-1} \text{ for } \mathrm{Re}(\lambda) \geqslant \lambda_0.$$

Hence $(\mathrm{Id} - B \cdot R(\lambda, A))^{-1}$ exists for $\mathrm{Re}(\lambda) \geqslant \lambda_0$. Since

$$(\lambda - (A + B))f = (\mathrm{Id} - BR(\lambda, A))(\lambda - A)f \quad \text{for all} \quad f \in D(A),$$

it follows that $(\lambda - (A + B))^{-1}$ exists and is given by

$$R(\lambda, A + B) = R(\lambda, A)(\mathrm{Id} - BR(\lambda, A))^{-1} \tag{2.1.10}$$

whenever $\mathrm{Re}(\lambda) \geqslant \lambda_0$. Now if A generates a holomorphic semigroup, there exists $M \geqslant 0$ such that $\|R(\lambda_0 + \mathrm{i}\eta, A)\| \leqslant M/|\eta|$ for all $\eta \in \mathbb{R}$. Consequently,

$$\|R(\lambda_0 + \mathrm{i}\eta, A + B)\| \leqslant \|(\mathrm{Id} - BR(\lambda_0 + \mathrm{i}\eta, A))^{-1}\| \cdot 2M/|\eta| \leqslant 2M/|\eta|$$

for all $\eta \in \mathbb{R}$. Thus $A + B$ generates a holomorphic semigroup by the corollary of Theorem 2.1.13. Moreover, it follows from Eq. (2.1.10) that $R(\lambda, A + B)$ is compact whenever $R(\lambda, A)$ is compact. Consequently, by Theorem 2.1.24 and the assertion proved above, $(S(t))_{t\geqslant 0}$ is compact whenever $(T(t))_{t\geqslant 0}$ is compact.

Finally assume that B is compact and $t_0 \geqslant 0$ such that $(T(t))_{t\geqslant 0}$ is norm continuous for $t > t_0$. Fix $t > t_0$. Denote by U the unit ball of E and fix $s \in (0, t]$. Then

$$\lim_{h\to 0} (T(t + s - h) - T(t - s))f = 0$$

for all $f \in \overline{BS(s)U} =: K$.

UNCORRECTED PROOF

Let $C = (A - \lambda)B(A - \lambda)^{-1} \in \mathcal{L}(E)$. Then $A + C$ is the generator of a strongly continuous semigroup by Theorem 2.1.28. We show that $A + B$ is similar to $A + C$. In fact, let $U = (\mathrm{Id} - BR(\lambda, A))$. Then U is an isomorphism on E such that $U(D(A)) = D(A)$. Moreover,

$$\begin{aligned} U(A + C)U^{-1} &= U(A - \lambda + C)U^{-1} + \lambda \\ &= U[(A - \lambda) - (A - \lambda)BR(\lambda, A)]U^{-1} + \lambda \\ &= U(A - \lambda)[\mathrm{Id} - BR(\lambda, A)]U^{-1} + \lambda \\ &= U(A - \lambda) + \lambda \\ &= A - \lambda + B + \lambda \\ &= A + B. \end{aligned}$$

□

Corollary 2.1.31 *Keeping the hypotheses and notations of Theorem 2.1.30 denote by $(S(t))_{t\geqslant 0}$ the semigroup generated by $A + B$.*

(i) *If $(T(t))_{t\geqslant 0}$ is norm continuous or compact or holomorphic, then $(S(t))_{t\geqslant 0}$ has the corresponding properties.*

(ii) *If B is compact as an operator on $D(A)$ endowed with the graph norm and if $(T(t))_{t\geqslant 0}$ is eventually norm continuous, then so is $(S(t))_{t\geqslant 0}$.*

Proof. Both assertions follow from Theorem 2.1.29 because $(US(t)U^{-1})_{t\geqslant 0}$ has $A + C$ as generator. □

2.1.9 Domains of Uniqueness

Given a semigroup $(T(t))_{t\geqslant 0}$, it is frequently difficult to determine the precise domain of its generator A. So it is important to know which (possibly strict) subspaces of $D(A)$ determine the semigroup uniquely. This can be formulated more precisely in the following way.

> *Let D_0 be a subspace of $D(A)$ and consider the restriction A_0 of A to D_0. Under which condition on D_0 is A the only extension of A_0 which is a generator?*

One obvious condition is that D_0 is a core. In fact, in that case, A is the closure of A_0. Since every generator B extending A_0 is closed, it follows that $A \subseteq B$ and hence $A = B$ since $\varrho(A) \cap \varrho(B) \neq \emptyset$.

We now show that cores are the only domains of uniqueness.

UNCORRECTED PROOF

The continuity of p implies that there exists a constant $c > 0$ such that

$$|p(f)| \leqslant c\|f\| \quad (f \in E). \tag{2.2.2}$$

Moreover, it follows from Eq. (2.2.1) on the preceding page

$$p(f) + p(-f) \geqslant p(0) = 0 \quad (f \in E).$$

A bounded operator T on E is called *p-contractive* if $p(Tf) \leqslant p(f)$ for all $f \in E$. Similarly, a semigroup $(T(t))_{t\geqslant 0}$ is called *p-contractive* if $T(t)$ is p-contractive for all $t \geqslant 0$. Of course, the most important case in this section is when p is the norm function N given by $N(f) = \|f\|$ $(f \in E)$. An N-contractive operator is just a contraction in the usual sense.

Remark However in Part-II, Chapter 2 and Part-III, Chapter 2 it will be important to dispose of a variety of sublinear functionals other than N. For example, we will consider N^+ on $C([0,1])$ given by

$$N^+(f) = \sup_{x\in[0,1]} f^+(x).$$

Then a bounded operator T is N^+-contractive if and only if T is positive and $\|T\| \leqslant 1$.

We first want to solve the following problem.

> *Given the generator A of a semigroup $(T(t))_{t\geqslant 0}$ find a condition on A which is equivalent to $T(t)$ being p-contractive for all $t \geqslant 0$.*

The *subdifferential* $\mathrm{d}p$ of p at f is defined by

$$\mathrm{d}p(f) = \{\varphi \in E' \colon \langle g, \varphi\rangle \leqslant p(g) \text{ for all } g \in E, \langle f, \varphi\rangle = p(f)\}. \tag{2.2.3}$$

It follows from the Hahn-Banach theorem that $\mathrm{d}p(f) \neq \emptyset$ for all $f \in E$.

Definition 2.2.1 An operator A on E is called *p-dissipative* if for all $f \in D(A)$ there exists $\varphi \in \mathrm{d}p(f)$ such that $\langle Af, \varphi\rangle \leqslant 0$ and A is called *strictly p-dissipative* if for all $f \in D(A)$ the inequality $\langle Af, \varphi\rangle \leqslant 0$ holds for all $\varphi \in \mathrm{d}p(f)$.

For convenience we want to have a distinctive name for the norm function. So we denote by $N \colon E \to \mathbb{R}$ the function given by $N(f) = \|f\|$ throughout. Then Eq. (2.2.3) can be written in the form

$$\mathrm{d}N(f) = \{\varphi \in E' \colon \|\varphi\| \leqslant 1, \langle f, \varphi\rangle = \|f\|\}. \tag{2.2.4}$$

A (strictly) N-dissipative operator is simply called (strictly) *dissipative*, which is in accordance with the usual nomenclature.

UNCORRECTED PROOF

Examples 2.2.2

(i) Let $E = C([0,1])$ and $f \in E$. Then there exists $x \in [0,1]$ such that $|f(x)| = \|f\|_\infty$. Define $\varphi \in E'$ by

$$\langle g, \varphi \rangle = \operatorname{sign}(f(x))\, g(x).$$

Then $\varphi \in \mathrm{d}N(f)$. Note that $\mathrm{d}N(f)$ may be an infinite set.

(ii) Let H be a Hilbert space and $0 \neq f \in H$. Then $\mathrm{d}N(f) = \{\varphi_f\}$ where

$$\langle g, \varphi_f \rangle = \|f\|^{-1}(g|f).$$

(iii) $A - \|A\|\,\mathrm{Id}$ is strictly dissipative for every bounded operator A.

Proposition 2.2.3 *Let A be an operator on E. Then A is p-dissipative if and only if*

$$p(f) \leqslant p(f - tAf) \text{ for all } f \in D(A) \text{ and } t > 0. \tag{2.2.5}$$

If in particular $(w, \infty) \subseteq \varrho(A)$ for some $w \in \mathbb{R}$, then A is p-dissipative if and only if

$$p(\lambda R(\lambda, A) f) \leqslant p(f) \text{ for all } f \in E, \lambda > w. \tag{2.2.6}$$

Proof. Assume that A is p-dissipative. Let $f \in D(A)$ and $t > 0$. There exists $\varphi \in \mathrm{d}p(f)$ such that $\langle Af, \varphi \rangle \leqslant 0$. Hence,

$$p(f) = \langle f, \varphi \rangle = \langle f - tAf + tAf, \varphi \rangle \leqslant \langle f - tAf, \varphi \rangle \leqslant p(f - tAf).$$

So Eq. (2.2.5) holds.

Conversely, let $f \in D(A)$. For every $t > 0$ choose $\varphi_t \in \mathrm{d}p(f - tAf)$. Then

$$\pm\langle g, \varphi_t \rangle \leqslant p(\pm g) \leqslant c\|g\|$$

for all $g \in E$, $t > 0$. Thus the net $(\varphi_t)_{t>0}$ is bounded. Consequently it posseses a $\sigma(E', E)$-limit point φ as $t \to 0$. We show that $\varphi \in \mathrm{d}p(f)$ and $\langle Af, \varphi \rangle \leqslant 0$.

Since $\langle g, \varphi_t \rangle \leqslant p(g)$ for all $t > 0$, it follows that $\langle g, \varphi \rangle \leqslant p(g)$ $(g \in E)$. Moreover,

$$\langle f, \varphi_t \rangle - t\langle Af, \varphi_t \rangle = p(f - tAf) \quad (t > 0).$$

Letting $t \to 0$ yields $\langle f, \varphi \rangle = p(f)$.

We have proved that $\varphi \in \mathrm{d}p(f)$. By hypothesis we have for all $t > 0$,

$$\begin{aligned} p(f) &\leqslant p(f - tAf) = \langle f - tAf, \varphi_t \rangle \\ &= \langle f, \varphi_t \rangle - t\langle Af, \varphi_t \rangle \leqslant p(f) - t\langle Af, \varphi_t \rangle. \end{aligned}$$

Consequently $\langle Af, \varphi_t \rangle \leqslant 0$ for all $t > 0$. Thus $\langle Af, \varphi \rangle \leqslant 0$. □

UNCORRECTED PROOF

Remark 2.2.8 Every half-norm p induces a closed proper cone

$$E_p := \{f \in E : p(-f) \leqslant 0\}$$

on E. Any p-contractive operator T on E leaves the cone E_p invariant (i.e., T is positive for the corresponding ordering).

Conversely, given a closed proper cone E_+ on E, then

$$p(f) := \operatorname{dist}(-f, E_+) = \inf\{\|f + g\| : g \in E_+\}$$

defines a half-norm on E such that $E_+ = E_p$. This half-norm is called the *canonical half-norm* on the ordered Banach space (E, E_+). The canonical half-norm is strict if and only if the cone E_+ is *normal* (this is equivalent to the fact that for every $\varphi \in E'$ there exist positive linear forms φ_1 and φ_2 on E such that $\varphi = \varphi_1 - \varphi_2$ (see Batty & Robinson [3] or Schaefer [37, Chapter V]).

Proposition 2.2.9 *Let A be a p-dissipative operator where p is a half-norm. If $D(A)$ is dense, then A is closable and the closure of A is p-dissipative as well (by Corollary 2.2.5).*

Proof. Let $(f_n)_{n\in\mathbb{N}}$ be a sequence in $D(A)$ with $\lim_{n\to\infty} f_n = 0$ and $\lim_{n\to\infty} Af_n = g$. We have to show that $g = 0$.

To this end let $h \in D(A)$. Then Eq. (2.2.5) on page 58 gives

$$p(f_n + th) \leqslant p(f_n + th - tA(f_n + th)) \quad (t > 0).$$

Letting $n \to \infty$ we obtain $p(th) \leqslant p(th - tg - t^2Ah) \quad (t > 0)$. Hence

$$p(h) \leqslant p((h - g) - tAh) \quad (t > 0)$$

by positive homogeneity. Letting $t \downarrow 0$ finally we obtain $p(h) \leqslant p(h - g)$ for all $h \in D(A)$. Since $D(A)$ is dense by hypothesis, we can approximate g by $h \in D(A)$ and conclude that $p(g) \leqslant p(0) = 0$. Since $\lim_{n\to\infty} A(-f_n) = -g$, we have $p(-g) \leqslant 0$ by symmetry. Hence $p(g) + p(-g) \leqslant 0$ which implies $g = 0$ by Eq. (2.2.9) on the preceding page. □

Lemma 2.2.10 *Let p be a half-norm and A a p-dissipative operator. Then*

$$\lambda\|f\|_p \leqslant \|(\lambda - A)f\|_p \text{ for all } f \in D(A), \lambda > 0. \tag{2.2.12}$$

In particular, $(\lambda - A)$ is injective for all $\lambda > 0$. If p is strict and A is closed, then $\operatorname{im}(\lambda - A)$ is closed for all $\lambda > 0$.

Proof. Let $\lambda > 0$, $f \in D(A)$. Then by Eq. (2.2.5) on page 58,

$$\lambda p(\pm f) \leqslant p((\lambda - A)(\pm f)).$$

UNCORRECTED PROOF